

Problem Set 4

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Data Sources I Would Like to Scrape

There are several data sources I would be interested in scraping for research purposes. First, the NBER Working Papers website (<https://www.nber.org/papers>) provides access to working papers in economics, which could be scraped to analyze trends in research topics, citations, and author networks over time. Second, EconJobMarket.org (<https://econjjobmarket.org>) contains job market candidate profiles and placement outcomes, which could be used to study patterns in PhD placement across institutions, fields, and demographic groups. Third, the IZA Discussion Papers database (<https://docs.iza.org>) which has a large collection of labor economics papers that could be scraped to study research trends in gender economics and discrimination. I would also be interested in scraping the World Bank Open Data API provides extensive development indicators for countries over time.

Q5(d): Object Types

The object `mydf` is of class `tbl_df`, `tbl`, and `data.frame` – that is, it is a tibble/dataframe. The object `mydf$date` is of class `character`, meaning it is stored as a character/string vector rather than a numeric or date type.

Q6(7): Spark vs. Tibble Classes

`df1` (a tibble) has class `tbl_df`, `tbl`, `data.frame`. `df` (a Spark dataframe) has class `tbl_spark`, `tbl_sql`, `tbl_lazy`, `tbl`.

Q6(8): Column Names

The column names differ between the tibble and the Spark dataframe. In the tibble, column names use dots (e.g., `Sepal.Length`), while in Spark they use underscores (e.g., `Sepal_Length`). This is because Spark/SQL does not allow dots in column names, so `sparklyr` automatically converts them.

Q6: Note on sparklyr

Despite attempting to install `sparklyr` on OSCER, the installation was unsuccessful due to dependency failures, particularly with the `dbplyr` package. This appears to be a compatibility issue between the version of R available on OSCER (R 4.0.2) and the current version of `sparklyr`. As noted in the problem set instructions, students would not be penalized if `sparklyr` does not work on the first try.